Lab 5 Files names permissions and storage on Linux

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Run on 9 November 2026

Purpose and required execution identity

The experiment connects pathnames, inodes, permissions and disk allocation using the original Lab 5 commands. Every command here ran on the Ubuntu x86_64 server as `oslab_igor`, UID 1001, without sudo or extra groups. An ordinary user's failures demonstrate permission checks that root may bypass. The complete recording is `Lab5-6/evidence/linux.txt`; the command sequence is `scripts/run-lab5.sh`.

```
+ whoami
oslab_igor
+ id
uid=1001(oslab_igor) gid=1001(oslab_igor) groups=1001(oslab_igor)
+ pwd
/tmp/igor-lab5.KuLBB2
```

Part 1 The tree and file types

The first letters in ls identified `/etc` and `/tmp` as directories (`d`), `/etc/passwd` as a regular file (`-`), `/bin` as a symbolic link (`l`), and `/dev/null` as a character device (`c`). `/bin` resolves to `/usr/bin`. File recognized ls as an ELF executable and passwd as text. A character device was `/dev/zfs`; a block device was `/dev/sr0`.

The accessible recursive totals were 476 MiB for `/var/log` and 3.5 GiB for `/usr`, making usr approximately seven to eight times larger. As requested, the du commands suppress permission diagnostics; inaccessible entries can make an ordinary user's total incomplete. These values describe this server at the time of the run.

```
+ ls -ld /etc /etc/passwd /bin /dev/null /tmp
lrwxrwxrwx   1 root root     7 Apr 22  2024 /bin -> usr/bin
crw-rw-rw-   1 root root  1, 3 Sep 28 13:52 /dev/null
drwxr-xr-x 119 root root 12288 Nov  9 14:10 /etc
-rw-r--r--   1 root root  1849 Nov  9 14:10 /etc/passwd
drwxrwxrwt  46 root root  4096 Nov  9 14:10 /tmp
+ ls -l /dev
+ head
total 0
crw-r--r-- 1 root root     10, 235 Sep 28 13:52 autofs
drwxr-xr-x 2 root root         300 Sep 28 13:52 block
drwxr-xr-x 2 root root          80 Sep 28 13:52 bsg
crw-rw---- 1 root disk     10, 234 Sep 28 13:52 btrfs-control
drwxr-xr-x 3 root root          60 Sep 28 13:52 bus
lrwxrwxrwx 1 root root           3 Sep 28 13:52 cdrom -> sr0
drwxr-xr-x 2 root root        3460 Sep 28 14:28 char
crw--w---- 1 root tty       5,   1 Sep 28 13:52 console
lrwxrwxrwx 1 root root          11 Sep 28 13:52 core -> /proc/kcore
+ file /usr/bin/ls /etc/passwd /bin
/usr/bin/ls: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked,
    interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=44900829
    b39271d878615772bb2a6ffa422f0891, for GNU/Linux 3.2.0, stripped
/etc/passwd: ASCII text
/bin:        symbolic link to usr/bin
+ readlink -f /bin
/usr/bin
+ find /dev -maxdepth 1 -type c -print
+ head -1
/dev/zfs
+ find /dev -maxdepth 1 -type b -print
+ head -1
/dev/sr0
+ du -sh /var/log
476M    /var/log
+ du -sh /usr
3.5G    /usr
```

Part 2 Inodes links and allocation

Stat showed inode 3113 and Links 1 for notes.txt. The inode number identifies the object; Links counts directory names referring to it. After ln, notes.txt and hard.txt both reported inode 3113 and two links. Removing notes.txt deleted one directory entry, leaving the data accessible through hard.txt with one link.

Soft.txt had a separate inode and an eight-byte length, exactly the length of its stored relative target `hard.txt`. After that target was deleted, soft.txt still existed but cat failed to resolve it. A symlink stores a pathname, not an additional hard link to the target inode.

```
+ ln notes.txt hard.txt
+ ls -li notes.txt hard.txt
3113 -rw-r--r-- 2 oslab_igor oslab_igor 6 Nov  9 14:10 hard.txt
3113 -rw-r--r-- 2 oslab_igor oslab_igor 6 Nov  9 14:10 notes.txt
+ rm notes.txt
+ cat hard.txt
hello
+ ls -li hard.txt
3113 -rw-r--r-- 1 oslab_igor oslab_igor 6 Nov  9 14:10 hard.txt
+ ln -s hard.txt soft.txt
+ ls -li hard.txt soft.txt
3113 -rw-r--r-- 1 oslab_igor oslab_igor 6 Nov  9 14:10 hard.txt
3115 lrwxrwxrwx 1 oslab_igor oslab_igor 8 Nov  9 14:10 soft.txt -> hard.txt
+ cat soft.txt
hello
+ rm hard.txt
+ cat soft.txt
cat: soft.txt: No such file or directory
+ printf 'broken symlink cat status=%d\n' 1
broken symlink cat status=1
+ ls -l soft.txt
lrwxrwxrwx 1 oslab_igor oslab_igor 8 Nov  9 14:10 soft.txt -> hard.txt
```

A new directory had two links: its entry in the parent and its own dot entry. The dot-dot entries in two new child directories added two references, making the parent's link count four. The two-byte one.txt consumed a 4 KiB allocation unit. Sparse big.img had logical length 1 GiB but no allocated data blocks, so du returned zero. File length and storage consumption measure different things.

Part 3 Permissions

The requested modes are 600 = `rw-------` for a private file; 644 = `rw-r--r--` for an ordinary shared-readable document; 755 = `rwxr-xr-x` for a public executable or traversable directory; and 700 = `rwx------` for a private executable or directory. Ownership selects the owner permission class; it does not override zero owner-read bits. Cat therefore failed at mode 000 even though the user owned s.txt. Restoring owner-read permission made the file readable.

Umask was 0022. Default file mode 0666 masked by 0022 became 0644, while default directory mode 0777 became 0755. This is bit masking: the mask removes requested permissions and cannot add execute bits to an ordinary new file.

```
+ chmod 600 s.txt
+ ls -l s.txt
-rw------- 1 oslab_igor oslab_igor 7 Nov  9 14:10 s.txt
+ chmod 000 s.txt
+ cat s.txt
cat: s.txt: Permission denied
+ printf 'mode 000 cat status=%d\n' 1
mode 000 cat status=1
+ chmod u+r s.txt
+ cat s.txt
secret
+ umask
0022
+ touch new.txt
+ mkdir newdir
+ ls -ld new.txt newdir
-rw-r--r-- 1 oslab_igor oslab_igor    0 Nov  9 14:10 new.txt
drwxr-xr-x 2 oslab_igor oslab_igor 4096 Nov  9 14:10 newdir
+ mkdir box
+ echo hi
+ chmod u=r,go= box
+ ls box
f
+ cat box/f
cat: box/f: Permission denied
+ printf 'read-only directory cat status=%d\n' 1
read-only directory cat status=1
+ chmod u=x,go= box
+ ls box
ls: cannot open directory 'box': Permission denied
+ printf 'execute-only directory ls status=%d\n' 2
execute-only directory ls status=2
+ cat box/f
hi
+ chmod 755 box
```

With only read permission on box, ls could read the name f but cat could not traverse the directory to open box/f. With only execute permission, ls could not list names but cat could access the already-known pathname. Directory r permits reading entries, while x permits searching and traversing them. These outcomes were actually observed under the unprivileged identity.

The s on `/usr/bin/passwd` is set-user-ID: its controlled code can execute with the file owner’s effective identity to update protected password information. The t on `/tmp` is the sticky bit, restricting removal of other users’ entries in a shared writable directory. Reading `/etc/shadow` was denied to oslab_igor, consistent with root/shadow ownership and its restricted mode. No password hashes were recorded.

Part 4 Disk diagram and proc

The disk diagram below shows sda, approximately 38.1 GiB, with sda1 as the ext4 root filesystem, sda14 as a 1 MiB boot-related partition without a displayed mount, and sda15 as a 256 MiB vfat EFI filesystem mounted at `/boot/efi`. Sr0 is a separate optical device. Root usage was 38 GiB total, 14 GiB used, 22 GiB available and 39 percent use; rounding and filesystem reservations explain why displayed columns need not sum exactly.

```
+ lsblk -o NAME,TYPE,SIZE,FSTYPE,MOUNTPOINTS
NAME    TYPE  SIZE FSTYPE MOUNTPOINTS
sda     disk 38.1G
|-sda1  part 37.9G ext4   /
|-sda14 part    1M
`-sda15 part  256M vfat   /boot/efi
sr0     rom  1024M
+ df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        38G   14G   22G  39% /
+ findmnt -no SOURCE,FSTYPE,TARGET /
/dev/sda1 ext4   /
```

Proc reported two processor entries and MemTotal 3,911,572 KiB, approximately 3.73 GiB. MemAvailable is an observation of the current workload, not installed capacity. Reading `/proc/self/status` with head reported Name head because self resolves for the process reading it. Likewise the fd listing described ls's own descriptors, including its open directory descriptor 3. Descriptors 1 and 2 point to the saved recording in this run because output was redirected there.

Conclusion

The outputs distinguish names from inode identity, symlink contents from target existence, and logical length from allocated space. The non-root identity makes the permission failures meaningful. Disk and proc values are observations of the actual Ubuntu server. Optional stretch exercises are not claimed as completed.

Execution captures and code excerpts

The native Terminal screenshot below was supplied by the author from a direct unprivileged permission test. The earlier full raw execution recording remains in the repository. This additional live check uses a separate temporary directory.

```
root@ubuntu-4gb-hel1-1:/tmp/igor-lab3-play-EN7rTn# runuser -u oslab_igor -- bash
oslab_igor@ubuntu-4gb-hel1-1:/tmp/igor-lab3-play-EN7rTn$ cd "$(mktemp -d /tmp/igor-lab5-live-XXXXXX)"
oslab_igor@ubuntu-4gb-hel1-1:/tmp/igor-lab5-live-7Ywq52$ whoami
oslab_igor
oslab_igor@ubuntu-4gb-hel1-1:/tmp/igor-lab5-live-7Ywq52$ printf 'permission test\n' > notes.txt
oslab_igor@ubuntu-4gb-hel1-1:/tmp/igor-lab5-live-7Ywq52$ chmod 000 notes.txt
oslab_igor@ubuntu-4gb-hel1-1:/tmp/igor-lab5-live-7Ywq52$ cat notes.txt
cat: notes.txt: Permission denied
oslab_igor@ubuntu-4gb-hel1-1:/tmp/igor-lab5-live-7Ywq52$ echo "read status: $?"
read status: 1
oslab_igor@ubuntu-4gb-hel1-1:/tmp/igor-lab5-live-7Ywq52$ chmod 600 notes.txt
oslab_igor@ubuntu-4gb-hel1-1:/tmp/igor-lab5-live-7Ywq52$ cat notes.txt
permission test
oslab_igor@ubuntu-4gb-hel1-1:/tmp/igor-lab5-live-7Ywq52$ umask 0022
oslab_igor@ubuntu-4gb-hel1-1:/tmp/igor-lab5-live-7Ywq52$ touch default.txt
oslab_igor@ubuntu-4gb-hel1-1:/tmp/igor-lab5-live-7Ywq52$ mkdir default_dir
oslab_igor@ubuntu-4gb-hel1-1:/tmp/igor-lab5-live-7Ywq52$ stat -c '%a %n' notes.txt default.txt default_dir
600 notes.txt
644 default.txt
755 default_dir
oslab_igor@ubuntu-4gb-hel1-1:/tmp/igor-lab5-live-7Ywq52$
```

Figure 1: Unprivileged permission checks

Lab 6 Inode identity and recursive pathname search

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New xv6 reproduction on 9 November 2026

Scope and implementation

This experiment examines two different tasks: reading metadata for a supplied name and walking a directory tree to discover names. The complete implementations are included as `xv6/user/mystat.c` and `xv6/user/find.c`, together with their UPROGS entries and the full kernel tree needed to build them. The following results were collected from that tree in QEMU with one emulated hart, using a fresh generated filesystem image.

The required README/r2 sequence was run first, including removing README and reading r2. README was then restored through a hard link so later trace tests could use it. An additional disposable file named specimen checks the behavior on a short payload. The test tree contains both file and directory matches.

Part 1 Types and empty directories

Types 1, 2 and 3 mean directory, regular file and device. Root has inode 1; dot and dot-dot both identify it because root is its own parent. A new directory a occupied 32 bytes, representing two 16-byte directory entries: a two-byte inode number and fourteen-byte name, for dot and dot-dot. The public stat structure exposes dev, ino, type, nlink and size.

```
$ mkdir a
$ ls a
.              1 35 32
..             1 1 1024
```

Reading metadata without confusing a name with an inode

`mystat` loops over its operands and calls `stat` for each one. It translates the three public xv6 types into file, directory, and device labels, and prints the inode number, link count, and byte size. If one lookup fails, it records failure and continues through later operands instead of aborting at the first bad name. The size uses the format for a 64-bit unsigned value rather than treating the field as a small integer.

README initially reported inode 2, one link and 2441 bytes. After ln, README and r2 shared inode 2 and two links. After removing README, r2 retained inode 2 and size 2441 with one link, and cat could still read its contents. The extra ten-byte specimen received inode 40 in this image and showed the same link behavior.

```
$ mystat README . console
README: type=file inode=2 links=1 bytes=2441
.: type=directory inode=1 links=4 bytes=1024
console: type=device inode=30 links=1 bytes=0
$ ln README r2
$ mystat README r2
README: type=file inode=2 links=2 bytes=2441
r2: type=file inode=2 links=2 bytes=2441
$ rm README
$ mystat r2
r2: type=file inode=2 links=1 bytes=2441
$ cat r2 | wc
48 336 2441


$ echo igor-file > specimen
$ mystat specimen
specimen: type=file inode=40 links=1 bytes=10
$ ln specimen alias
$ mystat specimen alias
specimen: type=file inode=40 links=2 bytes=10
alias: type=file inode=40 links=2 bytes=10
$ rm specimen
$ mystat alias
alias: type=file inode=40 links=1 bytes=10
$ cat alias
igor-file
$ mystat missing-path
mystat: cannot stat missing-path
```

Inode numbers depend on allocation order. The tested invariant is that both names identify one inode and that removing one name preserves the remaining link and contents.

Part 3 Measuring the limits

BSIZE is 1024, NDIRECT is 12, NINDIRECT is 1024 divided by a four-byte block number, and DIRSIZ is 14. Maximum file payload is `(12 + 1024/4) * 1024 = 274432` bytes, or 268 KiB. The indirect block is metadata, not part of that payload.

Creating averyveryverylongname stored the fourteen-character name averyveryveryl. The builder reported 2000 blocks: 47 metadata and 1953 data. Metadata comprises two boot/super blocks, 31 log blocks including its header, 13 inode blocks and one bitmap block.

```
42:#define IPB (BSIZE / sizeof(struct dinode))
5:#define BSIZE   1024 // block size
26:#define NDIRECT   12
27:#define NINDIRECT (BSIZE / sizeof(uint))
54:#define DIRSIZ 14
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total
    2000
$ mkdir averyveryverylongname
$ ls
averyveryveryl 1 36 32
[Other listing entries are retained in the full recording.]
```

The on-disk dinode fields are type, major, minor, nlink, size and twelve direct addresses plus one indirect address. A name is stored in a directory entry, not in the inode. Major and minor identify a device for a device inode.

Part 4 A find implementation that also reports directories

The search utility takes a starting pathname and a basename. After stat it compares the current name before deciding whether to recurse, allowing directories themselves to match. It copies the fixed-width `dirent.name` into a DIRSIZ-plus-one buffer and adds a terminator rather than assuming the kernel entry is already a C string.

The loop skips unused entries, dot, and dot-dot. Before constructing a child path, it checks the parent, separator, name, and terminator against its 512-byte buffer. Lookup and read failures produce diagnostics and a failure status. Every opened directory is closed after traversal. Ancestor descriptors remain open during recursion, so very deep trees are limited by xv6's descriptor limit.

Tests that distinguish traversal from correct matching

The tree contains `tree/target` and `tree/branch/target`, plus a directory `tree/branch/targetdir`. Searching for `target` printed the two files; searching for `targetdir` printed the directory; searching for `branch` printed the intermediate directory. A fourteen-character filename checked the DIRSIZ boundary, and a missing starting path exercised the diagnostic path. Invoking `find` without the two operands printed usage rather than interpreting uninitialized arguments.

```
$ mkdir tree
$ mkdir tree/branch
$ echo root-copy > tree/target
$ echo nested-copy > tree/branch/target
$ mkdir tree/branch/targetdir
$ find tree target
tree/branch/target
tree/target
$ find tree targetdir
tree/branch/targetdir
$ find tree branch
tree/branch
$ echo edge > tree/abcdefghijklmn
$ find tree abcdefghijklmn
tree/abcdefghijklmn
$ find missing-path target
find: cannot stat missing-path
$ find
usage: find start basename
$
```

These tests would expose a common bug in which basename comparison is performed only for regular files. Such a version can descend into `branch` and still fail to print `branch` or `targetdir`. The directory-target assertions in `scripts/run-xv6.py` verify that behavior explicitly. The raw console output is retained in `Lab5-6/evidence/xv6.txt`, so the displayed paths can be checked against the commands that constructed them.

Why unlink does not always free the blocks immediately

In `kernel/sysfile.c`, the `ip->nlink--;` line in `sys_unlink` decrements the link count after removing the name. In `kernel/fs.c`, `iput` waits for zero links and the final relevant in-memory reference before `itrunc` frees data blocks. An open reference can therefore preserve an unlinked inode.

The experiment proves survival through another hard link. The open-but-unlinked explanation comes from source inspection; it is a separate lifetime case.

Running `scripts/run-xv6.py` rebuilds the image and checks these distinctive link and search results. A fresh generated disk prevents leftover test names from affecting a rerun.

Execution captures and code excerpts

The following native Terminal screenshots were supplied by the author from direct source inspection and live xv6 execution. They replace the saved-record browser views. This additional run uses l6live_note and l6live_link (inode 48, 10 bytes, link counts 1 then 2 then 1), and a separate l6live directory tree. Earlier measurements in the report refer to the original specimen and tree fixtures retained in the raw transcript; inode numbers and directory link counts depend on filesystem state.

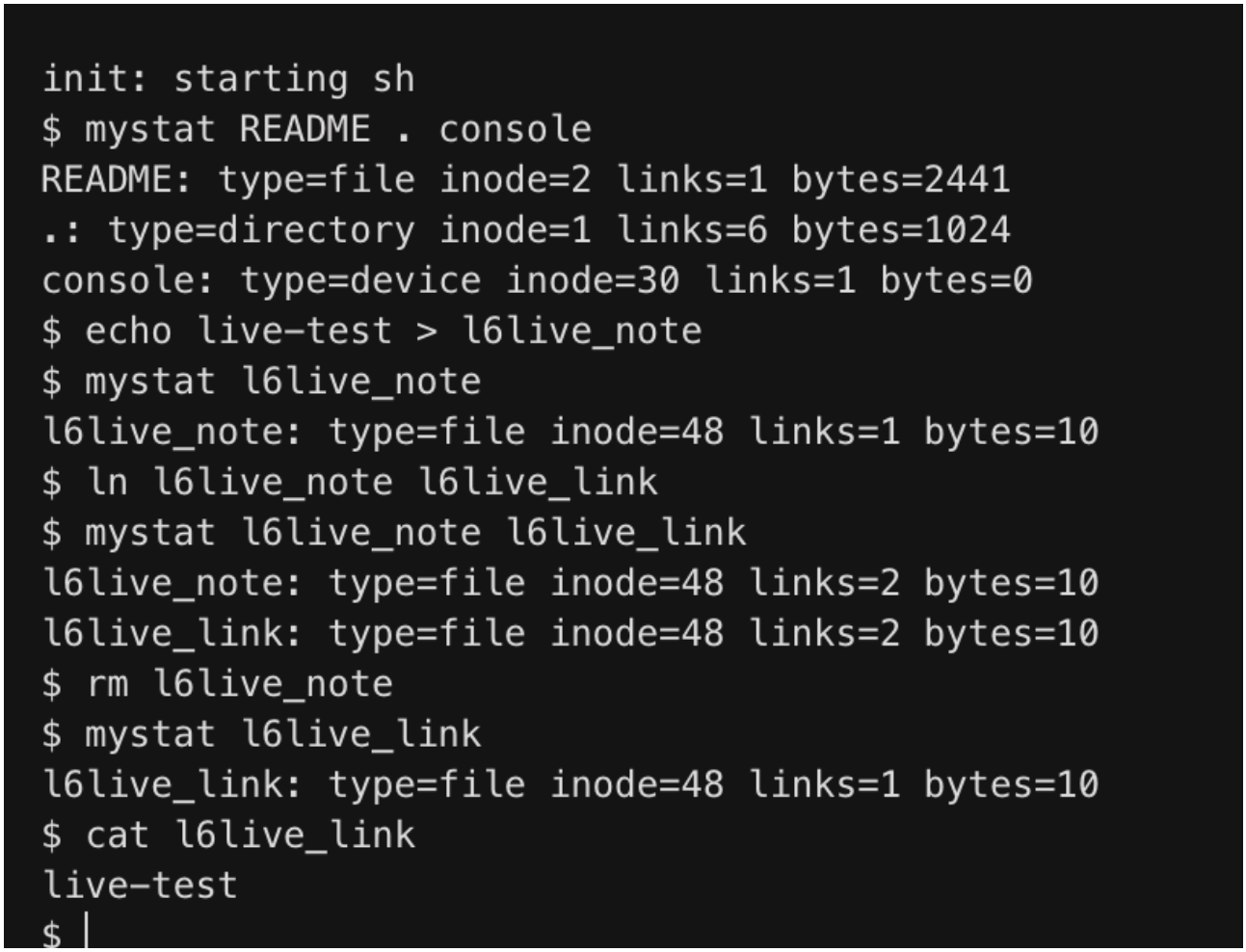


Figure 1: Native xv6 execution: metadata and hard links

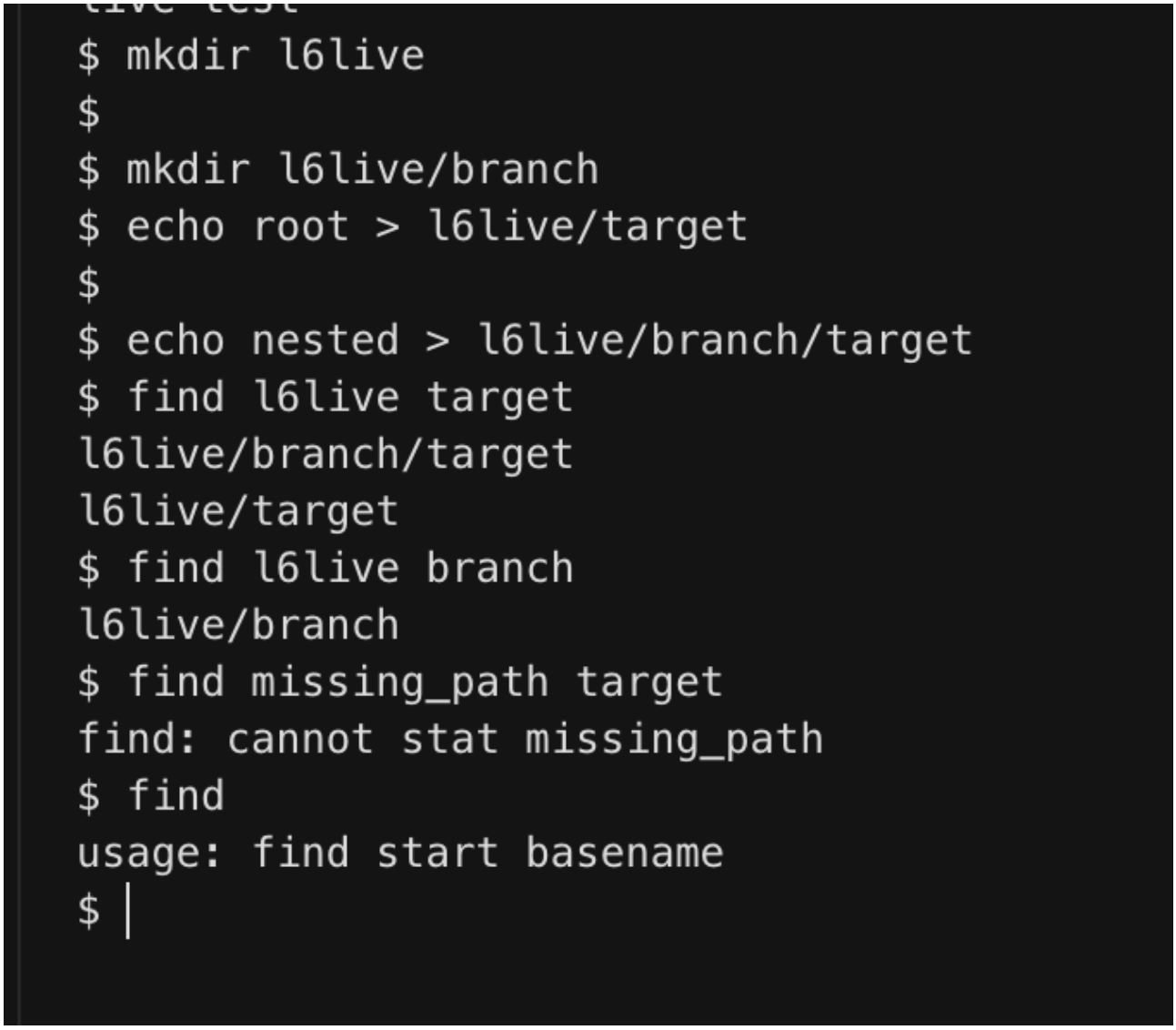


Figure 2: Native xv6 execution: recursive basename search and error paths

```
root@ubuntu-4gb-hel1-1:/tmp/igor-os-review-20261109/xv6# cat -n user/mystat.c
     1  #include "kernel/types.h"
     2  #include "kernel/stat.h"
     3  #include "user/user.h"
     4
     5  int
     6  main(int argc, char **argv)
     7  {
     8    if(argc < 2){
     9      fprintf(2, "usage: mystat path ...\n");
    10      exit(1);
    11    }
    12    int failed = 0;
    13    for(int i = 1; i < argc; i++){
    14      struct stat st;
    15      if(stat(argv[i], &st) < 0){
    16        fprintf(2, "mystat: cannot stat %s\n", argv[i]);
    17        failed = 1;
    18        continue;
    19      }
    20      char *kind = st.type == T_DIR ? "directory" :
    21                   st.type == T_FILE ? "file" :
    22                   st.type == T_DEVICE ? "device" : "unknown";
    23      printf("%s: type=%s inode=%d links=%d bytes=%lu\n",
    24             argv[i], kind, st.ino, st.nlink, st.size);
    25    }
    26    exit(failed);
    27  }
root@ubuntu-4gb-hel1-1:/tmp/igor-os-review-20261109/xv6# grep -n '_mystat' Makefile
137:    $U/_mystat\
root@ubuntu-4gb-hel1-1:/tmp/igor-os-review-20261109/xv6#
```

Figure 3: Native Terminal: mystat source and Makefile registration

```
root@ubuntu-4gb-hel1-1:/tmp/igor-os-review-20261109/xv6# sed -n '16,56p' user/find.c
static int
visit(char *path, char *target)
{
  struct stat st;
  if(stat(path, &st) < 0){
    fprintf(2, "find: cannot stat %s\n", path);
    return 1;
  }
  // A directory can match as well as a regular file.
  if(strcmp(basename(path), target) == 0) printf("%s\n", path);
  if(st.type != T_DIR) return 0;

  int fd = open(path, O_RDONLY);
  if(fd < 0){
    fprintf(2, "find: cannot open %s\n", path);
    return 1;
  }
  struct dirent de;
  char child[512], name[DIRSIZ + 1];
  int failed = 0, n;
  while((n = read(fd, &de, sizeof(de))) == sizeof(de)){
    if(de.inum == 0) continue;
    memmove(name, de.name, DIRSIZ);
    name[DIRSIZ] = 0;
    if(strcmp(name, ".") == 0 || strcmp(name, "..") == 0) continue;
    if(strlen(path) + strlen(name) + 2 > sizeof(child)){
      fprintf(2, "find: path too long under %s\n", path);
      failed = 1;
      continue;
    }
    strcpy(child, path);
    strcpy(child + strlen(child), "/");
    strcpy(child + strlen(child), name);
    failed |= visit(child, target);
  }
  if(n != 0){
    fprintf(2, "find: directory read failed for %s\n", path);
    failed = 1;
  }
  close(fd);
  return failed;
root@ubuntu-4gb-hel1-1:/tmp/igor-os-review-20261109/xv6# grep -n '_find' Makefile
136:    $U/_find\
root@ubuntu-4gb-hel1-1:/tmp/igor-os-review-20261109/xv6#
```

Figure 4: Native Terminal: find recursive traversal source and Makefile registration